

# File Reference: cloudflare/README.md

A file-specific explanation covering purpose, owner layer, metadata, source details, implementation signals, source excerpts, and safe-change rules.

| Item        | Explanation                                                                                    |
|-------------|------------------------------------------------------------------------------------------------|
| File        | cloudflare/README.md                                                                           |
| Category    | Documentation or text control file                                                             |
| Folder role | Cloudflare Worker/deployment area. Used for public AI backend and Cloudflare deployment notes. |
| Size        | 1,562 bytes                                                                                    |
| Extension   | .md                                                                                            |
| MIME type   | text/markdown                                                                                  |
| Text lines  | 47                                                                                             |

## Why this file exists

- Cloudflare setup notes for deploying the AI Worker endpoint and wiring the public site to the backend.

## How it is used

Cloudflare Worker/deployment area. Used for public AI backend and Cloudflare deployment notes.

## How this file participates in the system

This generated section reads the actual file and points out line numbers, declarations, endpoints, events, source excerpts, and debugging cues.

## API endpoints mentioned or called

| Item              | Explanation                              |
|-------------------|------------------------------------------|
| /api/portfolio-ai | Line 32. Handles AI assistant questions. |

## Important implementation signals

| Item                        | Explanation                                                                                                                                                                                                                                                                                                         |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cloudflare or AI at line 1  | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: # Portfolio AI Cloudflare Worker                                                                                                                                                                                             |
| Cloudflare or AI at line 6  | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: <a href="https://mauriceotieno.com/api/portfolio-ai">https://mauriceotieno.com/api/portfolio-ai</a>                                                                                                                          |
| Cloudflare or AI at line 9  | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: The static GitHub Pages site cannot safely hold API keys. The browser sends portfolio context to this Worker. The Worker uses OpenAI when `OPENAI_API_KEY` is configured, and otherwise falls back to Cloudflare Workers AI. |
| Cloudflare or AI at line 13 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: From this `cloudflare` folder:                                                                                                                                                                                               |
| Cloudflare or AI at line 16 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: npx wrangler login                                                                                                                                                                                                           |
| Cloudflare or AI at line 17 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: npx wrangler secret put OPENAI_API_KEY # optional, Workers AI works without it                                                                                                                                               |
| Cloudflare or AI at line 18 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: npx wrangler deploy                                                                                                                                                                                                          |
| Cloudflare or AI at line 21 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: `wrangler.toml` routes `/api/portfolio-ai` on `mauriceotieno.com` and `www.mauriceotieno.com` to this Worker.                                                                                                                |
| Cloudflare or AI at line 23 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: ## Cloudflare Requirements                                                                                                                                                                                                   |
| Cloudflare or AI at line 25 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: 1. The domain must be active in Cloudflare.                                                                                                                                                                                  |

| Item                        | Explanation                                                                                                                                                                                               |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cloudflare or AI at line 26 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: 2. DNS for `mauriceotieno.com` / `www` must be proxied through Cloudflare, not DNS-only, for Worker routes to run. |
| Cloudflare or AI at line 27 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: 3. The Worker must have the `AI` binding enabled. This repo's `wrangler.toml` includes it.                         |
| Cloudflare or AI at line 28 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: 4. Optional: add the `OPENAI_API_KEY` secret if you want the Worker to use OpenAI instead of Workers AI.           |
| Cloudflare or AI at line 32 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: <meta name="portfolio-ai-endpoint" content="/api/portfolio-ai" />                                                  |
| Cloudflare or AI at line 41 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: -Uri "https://mauriceotieno.com/api/portfolio-ai" `                                                                |
| Cloudflare or AI at line 47 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: Expected result: JSON with an `answer` field and a `provider` field such as `cloudflare-workers-ai` or `openai`.   |

## Representative opening snippet

```

1: # Portfolio AI Cloudflare Worker
2:
3: This Worker powers the public website endpoint:
4:
5: ```text
6: https://mauriceotieno.com/api/portfolio-ai
7: ```
8:
9: The static GitHub Pages site cannot safely hold API keys. The browser sends portfolio context to this
Worker. The Worker uses OpenAI when `OPENAI_API_KEY` is configured, and otherwise falls back to Cloudflare
Workers AI.
10:
11: ## Deploy
12:
13: From this `cloudflare` folder:
14:
15: ```powershell
16: npx wrangler login
17: npx wrangler secret put OPENAI_API_KEY # optional, Workers AI works without it
18: npx wrangler deploy
19: ```
20:
21: `wrangler.toml` routes `/api/portfolio-ai` on `mauriceotieno.com` and `www.mauriceotieno.com` to this
Worker.
22:
23: ## Cloudflare Requirements
24:
25: 1. The domain must be active in Cloudflare.
26: 2. DNS for `mauriceotieno.com` / `www` must be proxied through Cloudflare, not DNS-only, for Worker
routes to run.
27: 3. The Worker must have the `AI` binding enabled. This repo's `wrangler.toml` includes it.
28: 4. Optional: add the `OPENAI_API_KEY` secret if you want the Worker to use OpenAI instead of Workers AI.
29: 5. Keep the website meta tag:
30:
31: ```html
32: <meta name="portfolio-ai-endpoint" content="/api/portfolio-ai" />
33: ```
34:
35: ## Test
36:
37: After deployment:
38:
39: ```powershell
40: Invoke-RestMethod `
41:   -Uri "https://mauriceotieno.com/api/portfolio-ai" `
42:   -Method Post `
43:   -ContentType "application/json" `
44:   -Body '{"question":"Hi","context":{"profile":{"displayName":"Maurice Otieno"}}}'
45: ```
46:
47: Expected result: JSON with an `answer` field and a `provider` field such as `cloudflare-workers-ai` or
`openai`.

```

## Safe change checklist

- Confirm which app, website, installer, workflow, or document consumes this file.
- Change it on a feature branch from development.
- Regenerate docs if the file purpose, API surface, or behavior changes.
- Run the smallest relevant smoke test before merging into development.